25 Years of SMOTE: What Has Been Done and How Did It Benefit Society?

Data Section, Stand-Alone Draft

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This document typesets only the Data section of the survey with the JAIR template, so that the co-authors can read it in the journal's layout before it is merged into the full manuscript. Section numbering starts at 1 here and will change on merging; the reference list holds only the works this section cites.

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1 How the SMOTE Literature Uses Data

A reader who wants to compare two SMOTE variants, or to choose a benchmark for a new one, needs answers to questions that the papers rarely answer together. Can the numbers that two papers report on a dataset of the same name be compared? How broad is the evidence behind the benchmarks the field relies on? How difficult, beyond the imbalance ratio, are the datasets that can be checked? When a paper reports that its method beats SMOTE, on what kind of data does that happen, and can a reader check it? This section answers these questions from the 1,756 dataset entries and the comparisons with SMOTE that we extracted from the 76 papers of our corpus: 835 under one fixed metric per paper (834 per dataset and one average over datasets) and 2,497 over every metric the papers report. Three of them (ENN, AdaBoost and one review) contain no evaluation on data, so counts below use the 73 papers with an evaluation on data; Tomek links is kept because it illustrates its method on one synthetic two-class set.

The central observation is that the evidence for a new oversampler is only as interpretable as the description of its data, and in this literature that description is often incomplete: the same name can denote different tasks, many modifications are not described well enough to be repeated, and 14 of the 57 papers that use SMOTE as a baseline give no per-dataset pair of scores against it. Where per-dataset results exist, the proposed method beats plain SMOTE in about four of five decided comparisons. These wins are not more frequent on data the authors modified, but they are more frequent on data whose minority examples are borderline or isolated, and they are much less consistent on precision and specificity than on the other metrics.

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1.1 Extraction Procedure

For every paper we read the full text and any supplementary file in our collection, and recorded each dataset as printed, including spelling, with its group in the paper, source repository, size, number of features and classes, minority and majority counts, and the imbalance ratio where the paper prints them, together with the table and page they come from. For each dataset we also recorded how it reached the experiment: *original* (used as distributed, with its native classes), *repository-defined* (a two-class split published by a repository such as KEEL or imbalanced-learn, used as is), *modified* (the authors chose the minority classes, merged or dropped classes, set a threshold, undersampled or injected noise), or *synthetic* (generated by the authors). Where a paper did not state its source, two fixed rules were applied and flagged: names that follow the KEEL file convention (for example `ecoli-0-1_vs_5`, `yeast4`) were labelled repository-defined, and common names in multiclass papers with no stated change were labelled original. Two further rules were fixed before counting: a set that is two-class in its source and is used without comment was labelled original, and a set from which the authors removed rows (for example, records with missing values) or attributes was labelled modified. A set whose source the paper does not state, for which it prints no size, whose name identifies a public file and which the paper does not describe as changed was labelled original (four entries). One label rests on evidence outside the paper: the `toy` set of OREM-M, listed without a source, is the synthetic set of the ordinal-regression benchmark (Gutiérrez et al. 2016) and is labelled synthetic.

For the results we copied, per dataset, the printed score of the proposed method and of SMOTE from one results table selected by a fixed rule: G-mean, else AUC, else F-measure, else accuracy; the classifier highlighted in the abstract, else the first reported; the unperturbed condition. We recorded values rather than wins so that any definition of a win can be applied later. From the same table setting we then copied every other metric that the paper reports for both methods (2,497 dataset-by-metric values in 44 papers); the fixed-metric values were copied a second time in this pass, independently, and agree on all 839 rows that both copies contain. Plain SMOTE was distinguished from SMOTE embedded in a multiclass or ensemble wrapper (Static-SMOTE, OVO-SMOTE (Fernández, López, et al. 2013), SMOTE-all, SMOTE+AdaBoost). Values printed in a paper are never overwritten; corrections against the source repository are stored beside them with their justification. Every extracted value was then checked against the PDF a second time: a value was accepted when the dataset name and its numbers appear on the same line of the cited page, and every table that is printed as an image or whose text layer is garbled was read from the rendered page instead. All 1,654 dataset entries of the first extraction and 835 fixed-metric comparisons passed this check (the 93 sets added on 8 November were checked the same way, with a quoted sentence on the cited page); the values of the other metrics were read once, from the text layer or the rendered page.

1.2 The Benchmark Landscape

The 73 papers with an evaluation on data list 1,756 dataset entries, counting the 93 sets that ten papers generate only for an illustration or a side experiment (Section 1.7). One of them, the single toy example in Tomek links, is a description rather than a name and is set aside here, leaving 1,755 named entries. These reduce to 866 distinct names once names are lower-cased, text in parentheses is dropped, and underscores and spaces are treated alike. Treating spellings that differ only in punctuation as one dataset (`page-blocks`, `pageblocks` and the singular `page-block`) leaves 775 datasets, and grouping each numbered or split version with its main set (for example `yeast4` or `glass-0-1-6_vs_2` with yeast and glass), while keeping non-standard names such as `glass identification` apart, leaves 491 source families. Most datasets appear once: 552 of the 775 occur in a single paper (60 of them are the generated sets of a single experiment in MC-MBRC (Ma et al. 2024), which form one family). Counted as datasets, ecoli appears in 29 papers, yeast in 22 and glass in 19; counted with their splits, the ecoli family appears in 42 papers, the yeast family in 39 and the glass family in 35, about half of the 73. Half of all

uses of a family fall on 41 of the 491 families. Names whose printed number of features contradicts their main set are kept in its family when they are the same source with a different composition (for example `abalone8`, `oil1`) and kept apart when they are a different data set that shares the prefix (for example `car1`, a microarray set with 9,182 features).

Splits cut from one source are not copies of each other. Among the verifiable two-class KEEL files (Section 1.5), the 14 yeast splits range in imbalance ratio from 2.5 to 41.4 and in noise index from 0.29 to 0.86, and the 19 ecoli and 13 glass splits from 0.12 to 0.69 and from 0.24 to 0.77. They do share the source's attributes (7 or 8 for yeast, 6 or 7 for ecoli, 9 for glass), its domain and often its examples: `yeast4` and `yeast-0-5-6-7-9_vs_4` use the same 51 minority examples and differ only in which classes form the majority.

Outcomes on splits of one source are also related. In the eleven papers that compare with plain SMOTE on several splits of the same source (282 per-dataset comparisons, 262 same-family pairs; families by the rule in the evidence note), whether the proposed method wins does not cluster by family (pairs agree on win, tie or loss in 0.60 of cases, as under random relabelling), but the size of its margin does: the within-paper ranks of the margin differ by 0.29 on average for two splits of one source against 0.34 when family labels are permuted within each paper ($p = 0.005$, 10,000 permutations; leaving out one paper at a time gives p between 0.002 and 0.048).

Splits cut from one source vary the task (its imbalance ratio and where the minority examples lie) but not the source (its domain, attributes and dimensionality), and within the papers that evaluate several splits of one source, the margin over SMOTE tends to be slightly more alike within a family than between families. Papers should therefore report the number of source families alongside the number of datasets, and should not treat each split as independent evidence.

What the reader should do. Report the source family of each dataset alongside the count, so that readers can see how many independent sources stand behind a claim, and keep splits of one source together when testing for significance.

Candidate benchmarks for a cross-family comparison. Table 1 lists 18 datasets considered as benchmarks for comparing method families: the 11 datasets we set out to examine and the datasets that meet one of two written rules. The first rule takes every dataset used by at least ten papers (33 datasets). The second takes datasets used by three to nine papers whose every printed size is at least 10,000 instances, a size that no entry of the first group reaches (7 datasets). The groups follow from these two rules on use and printed size only and say nothing about overlap, noise, sparsity, feature type or any other property. IR is not used, because its definition differs between papers. No dataset used by three or more papers is printed with 100 or more features every time it is reported; this is a property of the rule, not evidence that the corpus contains no high-dimensional data sets.

Family counts follow the same rule for every dataset. The dataset `shuttle` is printed with four different sizes (2,175, 14,500, 43,500 and 57,999 instances) and once without a size. Two pairs of papers print matching sizes and class counts; the other versions are single reports, and OVO-SMOTE prints five classes in one table and seven class counts in another. The versions are therefore kept apart and not pooled.

The frequently used benchmarks are small: among the entries that print a size, half have 800 or fewer instances and half have nine or fewer features. Names printed with 10,000 or more instances in every paper are used by three to nine papers each, so a comparison across method families that wants to cover data size has to look beyond the frequent set.

Limits of the evidence. Size, dimensionality and the number of classes are printed for 75.6%, 72.4% and 78.3% of the 1,755 named entries, and class proportions in some form for 54.5%. A measured class overlap exists for 36 entries in two papers, on two incompatible scales, and the curated evidence contains no statement about sparsity. What the papers print therefore cannot show which ranges of overlap, noise or sparsity the benchmarks cover;

Table 1. Data Sets Considered as Benchmarks for Cross-Family Comparison. F: used by ten or more papers; D: used by three to nine papers, every printed size at least 10,000 instances; V: needs verification. Papers count the dataset (spellings that differ only in punctuation are one dataset); family papers include the numbered and split versions of the dataset. Values are as printed and differ between papers. [a]Four printed versions, not pooled.

Dataset	Group	Papers	Family papers	Instances	Size printed	Classes	Modified
ecoli	F	29	42	301–336	25/31	2, 4, 5, 8	8/31
page-blocks	F	22	34	545–5,476	17/22	2, 4, 5	6/22
yeast	F	22	39	483–1,485	16/22	2, 6, 8, 9, 10	4/22
pima	F	21	21	590–769	19/23	2	2/23
vehicle	F	21	30	713–940	19/23	2, 3, 4	10/23
haberman	F	20	20	252–306	19/22	2	2/22
glass	F	19	35	167–214	18/21	2, 6	7/21
balance	F	16	16	625	14/16	3	3/16
wine	F	16	16	130–178	15/18	2, 3	6/18
satimage	F	15	16	6,430–6,435	11/15	2, 6, 7	8/15
letter	D	9	9	11,057–20,000	9/9	2, 26	6/9
nursery	D	7	7	12,690–12,960	7/7	2, 4, 5	3/7
mammography	D	5	5	11,183	4/5	2	0/5
adult	D	3	3	32,562–48,842	2/3	2	1/3
avila	D	3	4	10,430–20,850	3/3	2, 11	1/3
default	D	3	3	30,000	2/3	2	0/3
shoppers	D	3	3	12,330–12,331	3/3	2	0/3
shuttle	V	7	15	2,175–57,999[a]	6/7	2, 5	1/7

the candidates in Table 1 are chosen on use and size only. Section 1.5 measures these properties directly for the entries whose files can be identified.

1.3 How Datasets Are Chosen and Modified

Table 2 shows that modification by the authors is common: 341 of 1,756 dataset entries were changed by the authors themselves, and for 195 the paper does not let a reader tell how the data were obtained. Eleven papers give no reason for their choice of datasets or only call them widely used.

Modification is not a fault in itself: binarising a multiclass set or raising the imbalance ratio are legitimate ways to probe a method. What matters for the reader is whether the modification is declared where a reader looks for it. A KEEL file name such as `yeast-0-5-6-7-9_vs_4` declares its split; a paper that prints only “Yeast” after choosing its own minority classes does not.

Modifications differ in how completely they are described. Of the 341 modified entries (43 papers), 177 in 30 papers change the class structure, most often by keeping one class against the rest (84 entries), and 147 in 9 papers raise the imbalance ratio on purpose, almost always by removing minority examples at random (131 entries). A purpose for the change is stated in 24 of the 43 papers. For 103 of the 341 entries, information needed to repeat the change is missing: the characteristics of the modified set are not printed for 65, the target imbalance ratio for 18 and the minority class for 15. At the paper level, 8 of the 37 papers that binarise data do not state which class becomes the minority. Entries modified without printed characteristics cannot be matched to any public file unless the authors publish the modified data themselves, as k-means SMOTE does; 48 such entries are therefore left out of the difficulty measurement (Section 1.5).

Table 2. How the Papers Choose and Obtain Their Data. Top: paper level, 73 papers with an evaluation on data. Bottom: dataset entries.

Item	Count
Reason given for the choice of datasets (papers)	
Own reason (diversity, difficulty, application, …)	48
Cites prior work	14
"Widely used" only	3
Not stated	8
Benchmark list adopted from earlier work, declared (papers)	20 (wholesale 11)
Binarised, minority rule not stated (papers)	8 of 37
How each dataset entry reached the experiment	
Original	537
Repository-defined (KEEL, imbalanced-learn)	470
Modified by the authors	341
Synthetic	213
Cannot be told from the paper	195

Table 3. Noise, Outliers and Missing Values Added to Real Data Sets. Entries: modified entries that record the altered versions (one per paper and kind of change).

Paper	Added	Levels	Data	Entries
SMOTE-IPF	class and attribute noise	20%, 40%	9 sets	2
RAMOBoost	class and attribute noise	several	abalone	2
RSMOTE	label noise; one outlier	10–30%; one point	all sets; fourclass	1; 1
RBO	label and feature noise	0–20% (label)	all sets	2
MC-CCR	label noise	0–25%	all sets	1
MC-MBRC	label noise	0–30%	real sets	1
MC-NCLWO	Gaussian feature noise	5–30%	4 sets	1
SPE	missing values	several	all sets	1

The remaining operations are named rather than pooled. Rows are removed or a subset is used in 18 entries (9 papers), features are removed, transformed or extracted in 15 (6 papers), three data sets are built and labelled by the authors from raw records (network traffic labelled by an intrusion detector, simulation output; 2 papers), and twelve carry label noise, attribute noise, an outlier or missing values added by the authors to real data sets (8 papers, Table 3). The noisy versions are recorded as one entry per paper and kind of change, since the papers print no size for them; the base sets keep their own category. Five further papers generate synthetic data with noise or outliers built in.

Why the data were modified can be read only where a paper says so. Across the 43 papers that modify data, ten state that the study or the method concerns two-class problems and convert multiclass sets for that reason, nine state that their data were not imbalanced enough and make them more imbalanced, eight add noise, outliers or missing values to test how the methods cope, two give practical reasons (the size of the data, agreement between the training and test files), and 19 give no reason that we could find. Counted by entries, 161 of the 341 modified entries are made harder in a way the paper explains (more imbalance or noise), 51 are converted for a two-class scope that the paper states, ten are changed for practical reasons and 127 have no stated reason. Which class

Table 4. Dataset Entries by Period of Publication and by How They Reached the Experiment (Percent of the Period's Entries). Synthetic: published by the authors / not published.

Period	Papers	Entries	Original	Repository	Modified	Synthetic	Cannot tell
up to 2004	5	36	33	0	58	0 / 0	8
2005–2009	9	110	44	0	26	0 / 26	4
2010–2014	8	154	40	0	36	0 / 20	4
2015–2019	18	592	18	35	25	9 / 1	11
2020–2024	24	618	36	28	12	2 / 12	10
2025–2026	8	245	33	37	5	0 / 2	22

becomes the minority, or which noise scheme is used, is traced to earlier work in five papers and to a written rule, such as the smallest class against the rest, in two.

Whether a modification favours the proposed method cannot be read from a paper, but some features of the design make it checkable. Of the 29 papers that convert classes, 16 evaluate on two-class tasks only, so that the conversion matches the scope of the method being tested; seven of them say so. Five of the 43 papers also use an unmodified version of a set they modify, which lets a reader see what the change does to the result. Twelve papers apply a change at several levels (imbalance ratios or noise levels), all of them papers that state that they make the data harder, and three publish their modified files. Where a paper reports comparisons with SMOTE both on sets it modified and on its other real sets (11 papers), the share of wins is higher on the modified sets in two papers, lower in six and equal in three (424 and 278 comparisons).

Modified versions are rarely shared. Of the 139 dataset families that someone modifies, 37 are modified in more than one paper; vehicle (11 papers), ecoli (10), abalone and satimage (9 each), glass and page-blocks (8 each) are modified most often, most of them by converting classes. The same modified name does not mean the same data: the nine papers that modify vehicle print seven different combinations of size and minority count, and no modified version is printed with the same size and minority count by more than three papers (vehicle 846/199, satimage 6,435/626, abalone 731/42, segment 2,310/330).

The way data reach the experiment has shifted over time (Table 4). Modified entries make up 58% of the entries before 2005 and 5% in 2025–2026, while repository-defined splits, absent before 2015, make up 28 to 37% of the entries since. Each period holds only five to 24 papers, so a single paper can move a share considerably. In the 32 papers from 2020 to 2026 the sets used most often are yeast (12 papers), new-thyroid and page-blocks (11 each) and ecoli (10) as distributed, mostly in multiclass papers, and the KEEL splits of yeast and glass (11 each), ecoli (9) and abalone (8). No modified set is used by more than four of these papers (page-blocks), and the synthetic sets that their authors publish (SVDD-WSMOTE, GMMSampling, OREM-M) are each used by one.

1.4 The Same Name Is Not the Same Data

Of the 192 datasets that at least two papers print with a size (spellings that differ only in punctuation counted as one dataset), 90 appear with more than one size: thyroid with 7,200 instances in five papers and 720 in four, dermatology with 358 (the KEEL version) in eight and 366 (UCI) in two, ecoli with 336, 327, 310 and 301, and pima with 768, 769, 635 and 590. Some of these differences are declared subsamples; the name alone does not tell which. MWMOTE (Barua et al. 2014) illustrates the problem inside a single paper: its Satimage (6,435 instances) and Statlandsat (4,435) are the full Statlog Landsat set and its training file, counted as two of its twenty datasets, and its common names are ambiguous in both directions (its Ecoli matches the KEEL file `ecoli1` exactly, while its Yeast matches no published split).

Table 5. Imbalance-Ratio Definitions Printed in the Papers

Definition	Formula	Used by (examples)
Largest over smallest class	$N_{\max}/N_{\min}$	MDO (Abdi and Hashemi 2016), CBHSID (Palli et al. 2022), MKC-SMOTE (Wang and Awang 2024) (its *IR*)
Majority over minority (binary wording)	$N_{\text{maj}}/N_{\min}$	OVO-SMOTE (defined for the binary case, applied to multiclass sets)
Percentage minority	$100\,N_{\min}/N$	RUSBoost (Seiffert et al. 2010)
Per-class imbalance degree	$\sum_{q\neq i} N_q/((C-1)N_i)$	OREM-M (Zhu, Liu, et al. 2023) (overall IR only cited, not printed)
Summed pairwise multiclass IR	$\sum_{i<j}(N_i/N_j - 1)$	SMOM (Zhu, Y. Lin, et al. 2017), GDHS (Yan et al. 2025), MKC-SMOTE (IR_{overall})
Imbalance degree *Id*	distance from uniform (Ortigosa-Hernández et al. 2017)	SHSampler (Zheng et al. 2022) (cited, no formula printed)
Minimum class prior	$N_{\min}/N$	DSRBF (Fernández-Navarro et al. 2011)
Coefficient of variation	$\text{sd}(N_i)/\text{mean}(N_i)$	COG-OS (Wu et al. 2007)
Proportion pair	e.g. 0.04:0.96	MWMOTE (wording: minority to majority)
Not stated (values equal $N_{\max}/N_{\min}$)	n/a	MC-CCR (Koziarski, Woźniak, et al. 2020), MC-RBO (Krawczyk et al. 2020)

Imbalance ratios add a second layer. Table 5 lists the definitions found in the corpus; some are not convertible into one another, and some papers print IR without a formula.

Errors also travel with shared benchmark tables: MC-CCR and MC-RBO both print an IR of 1.17 for vehicle, a value no definition produces from its class counts.

What the reader should do. Compare results across papers only after mapping each name to its exact file and class split, and report the IR formula together with its values.

1.5 How Difficult Are the Datasets That Can Be Checked?

The papers rarely describe how hard their data are beyond the imbalance ratio. A numerical overlap measure is printed for 36 of the 1,755 named entries, all in two papers, and noise or borderline examples are described for 5; three of 72 papers report such a property of their original data (Section 1.2). The literature on imbalanced learning, however, attributes much of the difficulty of these problems to characteristics of the data other than the imbalance ratio: overlap between the classes, noisy and borderline examples, small disjuncts and a shortage of minority examples (Jo and Japkowicz 2004; López et al. 2013; Napierala and Stefanowski 2016). We therefore measured these characteristics ourselves on the entries whose data can be identified. An entry counts as verifiable when its name matches a public file and the printed number of instances (and of minority examples, when printed) equals that file's, or when the paper prints no size, the name matches a public file and the authors did not modify the data, or when the authors publish the modified file themselves under the entry's name (k-means SMOTE (Douzas, Bacao, and Last 2018)). One paper (Loyalty-SMOTE) prints, for eight entries, a size exactly one larger than the public file, apparently counting the header line; for this paper a printed size of the file's size plus one is accepted. A two-class file is not accepted for an entry whose paper prints more than two classes. The files come from KEEL (Alcalá-Fdez et al. 2011), the UCI repository (Kelly et al. 2026) (downloaded from the repository, and through the UCI++ collection (Garcia n.d.), the UCI files distributed with Weka and single-file copies), LIBSVM (Chang

Table 6. Difficulty Profile of the Verifiable Tasks. A task can carry several flags; entries count how often the corpus uses a task.

	Binary and binarised		One-vs-rest
	Tasks	Entries	Tasks
All	238	703	347
Small disjuncts (rare ≥ 20%)	33 (14%)	65 (9%)	29 (8%)
Outliers (> 20%)	83 (35%)	198 (28%)	49 (14%)
Small sample ($N_{\min}/d < 10$)	160 (67%)	454 (65%)	162 (47%)
None of the three	65 (27%)	224 (32%)	174 (50%)
Imbalance ratio > 9	140	376	164
Noise index, median [P25, P75]	0.57 [0.29, 0.77]		0.29 [0.07, 0.61]

and C.-J. Lin 2011), PMLB (Romano et al. 2022), imbalanced-learn (Lemaître et al. 2017), the PROMISE defect data (Jureczko and Madeyski 2010) and the repositories in which five papers publish their own processed data, Kaggle for the sets that papers take from it, and the CLUTO clustering toolkit's site for two text collections. A name is looked up under the other published names of the same data set (liver disorders is `bupa`, contraceptive method choice is `cmc`), but only the printed size decides whether a file is accepted. This holds for 991 entries from 65 papers, which use 261 files. For 133 of these entries (15 papers) the paper prints no size and describes no change to the data, and for 36 of them it prints no class proportions either; their label and their measured profile rest on the assumption that the public file was used as distributed. Entries modified by their authors without printed characteristics and without a published file (48) are left out, because the data actually used cannot be reconstructed.

Each two-class file, each binarisation that a paper describes precisely enough to reproduce, and each imbalanced-learn task defines one task: 238 tasks, used by 703 entries in 46 papers. One-vs-rest tasks that we built from multiclass files (347) are not tasks any paper evaluated and are reported apart. Every minority example is labelled safe, borderline, rare or outlier from the classes of its five nearest neighbours (Napierala and Stefanowski 2016), with distances measured by HVDM (Wilson and Martinez 1997). The noise index of a task is the share of minority examples that are borderline or outliers; we report it as a continuous value because we found no published threshold for it. Three flags use published thresholds: small disjuncts when at least 20% of minority examples are rare and outliers when more than 20% are outliers (Napierala and Stefanowski 2016), and small sample when there are fewer than ten minority examples per feature (Jain et al. 2000). The imbalance ratio $N_{\text{maj}}/N_{\min}$, computed from the files, is grouped into the bands 1.5 to 3, 3 to 9 and above 9 (Fernández, García, et al. 2008) and is not used as a difficulty flag.

Table 6 shows that the most common property of the binary tasks is a small sample: 160 of 238 have fewer than ten minority examples per feature. The flags combine in seven profiles: small sample alone (87 tasks), small sample with outliers (42), all three flags (28), outliers alone (11), small disjuncts with a small sample (3), small disjuncts with outliers (2) and none (65). The noise index separates tasks that the flags do not (Figure 1a): among the 65 tasks without a flag it ranges from 0 to 0.84, with 21 tasks at 0.5 or above (haberman 0.84, the numeric German credit data 0.81, contraceptive 0.79 and bupa 0.79 among them) and 16 below 0.1. Its median is 0.31, 0.44, 0.50 and 0.65 in the imbalance-ratio bands up to 1.5 (6 tasks), 1.5 to 3 (40), 3 to 9 (52) and above 9 (140), and its rank correlation with the imbalance ratio is 0.25.

The measurable part of the corpus is also low-dimensional (Figure 1b). The printed number of features has a median of 9 and half of the entries lie between 4 and 10 features. The 59 entries with more than 100 features come from 54 names in 18 papers, no name is used by more than two papers, and only eight of them could be

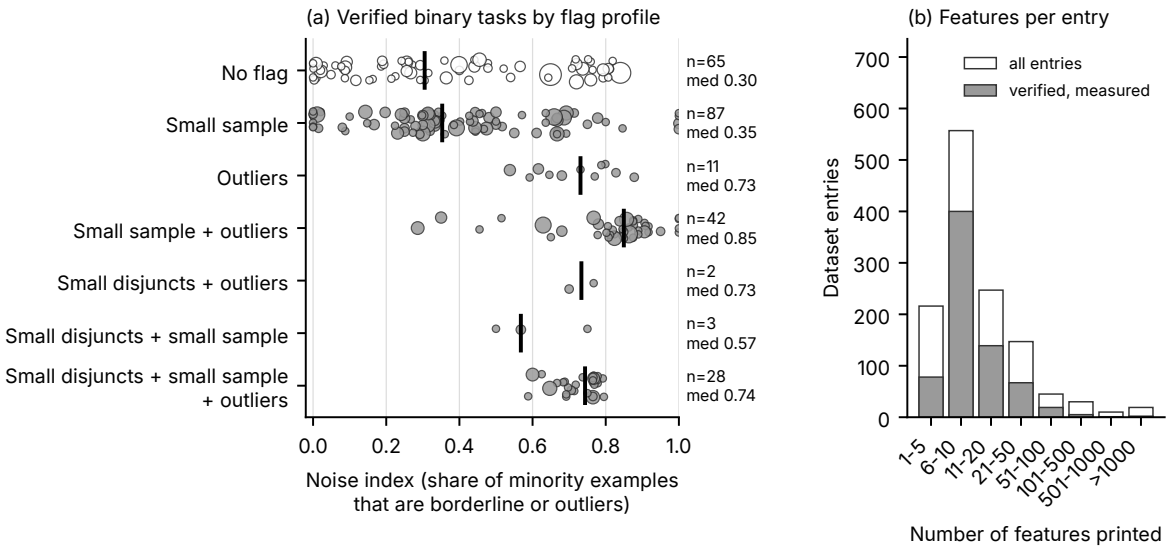


Fig. 1. (a) Noise index of the 238 verifiable binary and binarised tasks, one row per flag profile; marker area grows with the number of entries that use the task, and the vertical bar marks the median. (b) Number of features printed for the 1,271 entries that report it, with the entries that could be verified and measured. The bins are for display only.

verified (musk twice, isolet, madelon, a1a and mushrooms in the files that HDE publishes, and the text collections k1b and la12 of COG-OS); the verified two-class tasks have at most 617 features (median 9), and k1b, measured as one-vs-rest tasks, has 21,839.

The imbalance ratio, the one difficulty-related number most papers print, says little about where the minority examples lie: tasks with a low ratio can be as mixed as tasks with a high one, and a task without any flag is not necessarily easy. Conclusions drawn from the common benchmarks speak mainly for small, low-dimensional tables with few minority examples per feature.

Most papers mix these profiles. Of the 40 papers with at least three verifiable tasks, 37 include both flagged and flag-free tasks, two (COG-OS (Wu et al. 2007) and SMOTE-ENN) use only flag-free tasks and one (GOIO) only flagged ones. Section 1.6 relates the measured profiles to the reported wins.

A reference collection. For readers who need a small benchmark that covers the measured range, Table 7 selects 42 of the 238 tasks by a fixed rule with no threshold of our own. Each combination of flag profile and imbalance-ratio band that occurs (19 combinations) contributes the task that the corpus uses most, for comparability with earlier results, and the tasks with the lowest and the highest noise index, for spread. The collection spans noise indices from 0 to 1, imbalance ratios from 1.1 to 970.6 and sizes from 79 to 62,630 instances, and 13 of its files match the official KEEL table. Its tasks are not independent of one another: it holds four yeast splits, three pima versions and two each of ecoli, glass, haberman, cleveland, heart and iris, which should be grouped by source when results are tested.

Limits of the measurement. Part of the files are copies of the repositories rather than downloads from the repositories themselves: the keel-ds and imbalanced-databases packages, two independent GitHub copies of the KEEL standard collection, PMLB, UCI++, the UCI files distributed with Weka, a copy of the tera-PROMISE defect

data and single UCI files from a public data-set mirror. The UCI and LIBSVM files added in the second pass were downloaded from the repositories, and five papers' own files from the repositories their authors name (HDE, DGOT, SOS, MBS and the k-means SMOTE release). Eleven Kaggle data sets that papers name were downloaded from their Kaggle pages, and two text collections from the CLUTO toolkit's site; Kaggle files are uploaded by users, and a file may differ from the one a paper used. For the 72 KEEL files in the official KEEL table, instance and feature counts match in every case, and the two copies of the KEEL standard collection agree in instances and features on all 67 files they share; the other files have not been compared with the repositories. When several copies hold the same data set, each entry added in the second pass is mapped to one copy, the authors' own files and KEEL first and PMLB last, because PMLB stores nominal attributes as integer codes, which then enter HVDM as numeric attributes. Rows with missing values are dropped before the neighbours are found; the largest losses are 1,129 of the 3,772 rows of the sick thyroid data and 75 of the 155 rows of hepatitis. The labels are computed in the full feature space, not in any two-dimensional projection. Two verifiable files were not measured: covertype (581,012 rows), which two entries use only as a seven-class file, so that it would give one-vs-rest tasks only, was beyond our computing time, and la12 (6,279 documents, 31,472 term features), which one entry uses as a six-class file, was beyond the memory of our machine. Of the 866 dataset names, 159 have no file in these copies (45 of them name a public source such as UCI or KEEL; the others are noisy versions made in an experiment, private data, sites that no longer exist, image, text and time-series collections, or names that do not identify a data set), 196 are synthetic and 82 have a file whose size differs from every printed one.

1.6 Where Reported Wins over SMOTE Come From

At the paper level, 44 of the 73 papers use plain SMOTE as a baseline, 13 use SMOTE inside a multiclass or ensemble scheme, and 16 do not compare with SMOTE at all. Fewer papers give scores a reader can pair: 29 papers print per-dataset scores for the proposed method and plain SMOTE in the same table. For these papers we compared the printed values on every metric that the table reports for both methods, with the classifier and condition fixed as in Section 1.1; a win means the proposed method's printed value is better in the metric's direction, and equal printed values count as a tie. This gives 1,511 comparisons on 524 datasets (one to six metrics per paper, median three); 16 cannot be decided because a value is not printed as a number. Of the other 1,495 (520 datasets), the proposed method wins 1,107, ties 102 and loses 286, a win in 79.5% of the decided comparisons (95% CI 77.3 to 81.5). Twenty-three of the 29 papers report at least one comparison that SMOTE wins.

Table 8 shows that the margin is not the same on every metric. On AUC, G-mean, F-measure and recall the proposed method wins 76 to 83% of the decided comparisons, and on accuracy and MCC 91 and 83%; on precision the split is only 62 to 46, and on specificity SMOTE wins six of seven. Counting one result per dataset (the majority over its metrics) gives 409 wins, 35 ties and 76 losses on 520 datasets, and the earlier single-metric rule (G-mean, else AUC, else F-measure, else accuracy) gives 521 comparisons, in which the proposed method wins in 402 rows, ties in 39 and loses in 80.

We tested the hypothesis that motivated this analysis: that proposed methods win more often on data the authors modified or generated than on original data. The data do not support it (Table 9). Modified and synthetic data give a win in 75.7% of decided comparisons (373 of 493) against 79.4% on original data. Because the comparisons of one paper are not independent, we estimated the difference with the paper as the unit of clustering: a logistic model with paper-level clustering (generalised estimating equations) gives an odds ratio of 0.69 (95% CI 0.43 to 1.08, $p = 0.10$) for modified or synthetic against original data; a conditional logistic model that compares the two kinds of data only inside the 12 papers that use both and report both outcomes gives 0.62 (0.39 to 0.99, $p = 0.04$); and a mixed model with random effects for paper and for dataset gives 0.75 (0.58 to 0.98). The direction is therefore the opposite of the hypothesis, but it is not stable: with one result per dataset the two clustered estimates are 0.66 ($p = 0.19$) and 0.51 ($p = 0.17$), and under the single-metric rule 0.83 ($p = 0.58$) and 0.69 ($p = 0.43$). Paper by paper,

among the 14 papers with both kinds of data, the win share is higher on modified or synthetic data in 4, higher on standard data in 7 and equal in 3 (sign test $p = 0.55$). Repository-defined splits do not differ from original data (odds ratio 1.19, $p = 0.61$), and the comparisons whose data origin cannot be told show the highest raw win share apart from the two papers with synthetic data, a difference that disappears once papers are taken into account (odds ratio 0.99, $p = 0.98$).

The second hypothesis concerned papers that add a single standard dataset to data of another kind. Two papers fit this pattern: MWMOTE (one standard and 19 other datasets) and RUSBoost (Seiffert et al. 2010) (one and 14). Both win on their standard dataset; on their modified datasets and those of unstated origin they win 52 of 65 decided comparisons (80.0%; two MWMOTE datasets that could not be matched to a record are left out), the same share as on such data in the 22 papers that have at least two standard datasets and also report other data (484 of 603, 80.3%). Four papers compare with SMOTE only on standard data and one (INOS) only on modified data. Two papers are too few for a test, and nothing in them suggests that the standard dataset serves a different purpose from the rest.

Measured difficulty says more about where the wins fall than the origin of the data. For the 107 verifiable two-class files (Section 1.5) that appear in these comparisons (23 papers, 496 comparisons), the chance that the proposed method beats SMOTE rises with the noise index: the odds ratio per 0.1 of noise index is 1.15 (95% CI 1.04 to 1.26, $p = 0.005$) with paper-level clustering and 1.19 (1.05 to 1.34, $p = 0.006$) within papers, and 1.21 (1.06 to 1.37, $p = 0.004$) with one result per file and paper. The imbalance ratio shows no such relation (odds ratio 0.93 per doubling, $p = 0.62$), nor do the small-sample, outlier and small-disjunct flags taken one at a time. This analysis is exploratory: it covers the files that could be verified, which are mostly small and low-dimensional.

Comparisons with SMOTE inside another scheme are kept apart and are not pooled with Tables 8 and 9. Twelve papers contribute 558 comparisons: Static-SMOTE (EMDO (Yao and T.-B. Lin 2021), OVO-SMOTE, SOUP) 79, with 69 wins, 3 ties and 7 losses; SMOTE after one-versus-one decomposition (AMDO (Yang et al. 2018), GDHS (Yan et al. 2025), SHSampler (Zheng et al. 2022)) 148, with 101, 6 and 41; SMOTE applied to all classes (MC-CCR (Koziarski, Woźniak, et al. 2020), MC-RBO (Krawczyk et al. 2020), NROMM) 239, with 231, 0 and 8; and SMOTE combined with another learner (DSRBF (Fernández-Navarro et al. 2011), EasyEnsemble, GAMO) 92, with 71, 2 and 19. Three papers are left out of every comparison set. KNNOR (Islam et al. 2022) prints only means over its 19 datasets (G-score .8363 for KNNOR and .8238 for SMOTE). MC-MBRC (Ma et al. 2024) names SMOTE as a baseline, but the column headers of its per-dataset tables do not say which column is SMOTE. In RBBoost (Díez-Pastor et al. 2015), the scores of its RB-Boost come from the Boosting-family table of its supplement (Supplementary Table 11) and those of its SMOTE ensemble (ESM) from the data-processing-family table (Supplementary Table 3); both report the same protocol (J48 with Laplace smoothing, ensembles of 100, stratified 5×2-fold cross-validation), but the paper compares methods only within each family and never compares RB-Boost with ESM directly. Its 86 datasets, 63 of them repository-defined, are kept in our data for audit.

What the reader should do. Read a reported win over SMOTE together with the metric: precision and specificity often favour SMOTE even when the other metrics do not. Expect wins over SMOTE to be more frequent on data whose minority examples are borderline or isolated, and less informative on clean data. Whether the data were modified does not, on this evidence, inflate the win rate; whether they are described well enough to be identified matters more for the reader (Sections 1.4 and 1.7).

1.7 Can the Reported Results Be Checked?

Of the 57 papers that use SMOTE as a baseline (44 plain, 13 wrapped), 13 give a reader no per-dataset pair of scores in the paper itself: the results are plots (Borderline-SMOTE (H. Han et al. 2005), Safe-Level-SMOTE (Bunkhumpornpat et al. 2009), Cluster-SMOTE (D. Cieslak et al. 2006), k-means SMOTE (Douzas, Bacao, and Last 2018), GLOS (M. Han et al. 2023)), averages (G-SMOTE (Douzas and Bacao 2019), CISO (Yadav and Vijay 2025), SplitBal (Sun

et al. 2015), QC-SMOTE (Upman and Gowda 2026)), or ranks (MDO, RBO (Koziarski, Krawczyk, et al. 2019), MC-NCLWO (Tao et al. 2025)), and the review (Hassanat et al. 2022) compares synthetic points rather than classifiers. A fourteenth paper, KNNOR, prints its pair of scores only as a mean over its datasets.

Even where per-dataset tables exist, they do not always cover the datasets a paper lists. In the 43 papers with per-dataset comparisons with SMOTE, 201 of the 1,034 listed dataset entries (19.4%, 95% CI 17.1 to 22.0) have no row in the table that compares the proposed method with SMOTE. Seventy-six of them are sets that seven of these papers generate only for an illustration or a side experiment (described below with the other synthetic data), which are not meant to be compared, and six are noisy versions of real sets that five papers use only in a separate robustness experiment. Of the other 952 entries, 119 (12.5%, 95% CI 10.5 to 14.8) have no such row, and 12 papers account for all of them, most of all SMOTE-IPF (Sáez et al. 2015) (32 of 41) and Loyalty-SMOTE (30 of 59). The gap is largest for synthetic sets (38 of 48) and for entries whose origin cannot be told (44 of 166), and smallest for modified and repository-defined data (5 of 180 and 7 of 189). Some of these datasets appear in other results tables of the same paper; what is missing for them is the pair of scores against SMOTE.

Synthetic data are the hardest to check, because a reader can only regenerate them. In 29 papers, 213 dataset entries are generated data; 93 of them are sets that ten papers generate only for an illustration or a side experiment, which we count as entries as well. Seven papers give a location for those data (63 entries; for example k-means SMOTE (Douzas, Bacao, and Last 2018) and G-SMOTE (Douzas and Bacao 2019)), three take them from cited work (13 entries; NROMM reuses the four simulated sets of SVDD-WSMOTE), MC-MBRC (Ma et al. 2024) states that its 60 generated sets are available on request, RSMOTE names only the generator (3 entries), five link only the code of their method (10 entries), and thirteen say nothing about where the data can be obtained (64 entries, among them the 24 sets of SPIDER2). OREM-M is counted twice: its simulated illustration has no location, while its `toy` set, which the paper lists without a source, is the synthetic set of the ordinal-regression benchmark (Gutiérrez et al. 2016); the public copy has the same 300 examples, two features and five class sizes, so it is an unmodified published set.

Baseline results are also reused across papers. QC-SMOTE copies its baseline numbers from the VS-SMOTE paper (Qiu et al. 2025), and every SMOTE-all value in MC-CCR's Table 6 is identical, to two decimals and on all 20 datasets, to the SMOTE-all column of MC-RBO's Table VI, from the same group and with the same C5.0 setup. The same pattern appears across research groups: the Baseline, SSMOTE, GCS, ABNC, OSMOTE and AMDO columns of EMDO's Table 5 (Yao and T.-B. Lin 2021) repeat, value for value on all nine shared datasets, the MAUC results of AMDO's Table 6 (Yang et al. 2018); EMDO states that it uses the same datasets as AMDO but not that the baseline results were copied. Reusing a baseline is efficient, but unless it is stated, a reader cannot tell whether the comparison was run under identical conditions. Several of these papers place per-dataset results on GitHub or in material outside the paper. Where only ranks are given, the paper's own table can still be informative: in RBO's Table 2 (CART, no added noise) SMOTE has the better mean G-mean rank, and CISO reports itself statistically equivalent to SMOTE.

For these papers the headline claim cannot be falsified from what is published. This does not mean the claims are wrong; it means a reader cannot tell whether an average improvement comes from consistent gains or from large gains on a few datasets.

1.8 Recommendations for Reporting Data

(1) Name the repository and version of every dataset, and whether the full set or a training file was used (Section 1.4).
(2) State which class or classes form the minority and why, in the text and not only in a table (Section 1.3).
(3) Declare every modification (binarisation, class merging, undersampling, noise, dropped attributes) in the data section, and print the size and class counts of the data actually used (Section 1.3).

(4) Give the imbalance-ratio formula, not only its values (Section 1.4).
(5) Report the number of source families alongside the number of datasets, and keep splits of one source together when testing for significance (Section 1.2).
(6) Report per-dataset results for the proposed method and plain SMOTE on every metric used, with standard and modified datasets shown separately (Sections 1.6 and 1.7).
(7) Separate tuning datasets from evaluation datasets, as CISO and LS (D. A. Cieslak and Chawla 2008) do, and run the baselines rather than copying their numbers (Section 1.7).
(8) Report at least one measure of where the minority examples lie (for example the shares of safe, borderline, rare and outlier examples) next to the imbalance ratio (Sections 1.5 and 1.6).

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Table 7. A Reference Collection of Verifiable Two-Class Tasks. One row per selected task; Role: core = most used task in its cell, low/high = lowest/highest noise index in its cell. Profile: S small sample, O outliers, D small disjuncts; Noise: noise index; Entries: corpus entries that use the task.

Task	Role	Profile	IR band	N	$N_{\min}$	d	IR	Noise	Entries
banknote-authentication	low	none	≤1.5	1,372	610	4	1.2	0.00	1
bupa	core+high	none	≤1.5	345	145	6	1.4	0.79	6
smoke detection	low	none	1.5–3	62,630	17,873	14	2.5	0.00	1
pima	core	none	1.5–3	768	268	8	1.9	0.65	24
haberman	high	none	1.5–3	306	81	3	2.8	0.84	22
segment0	low	none	3–9	2,308	329	19	6.0	0.02	7
yeast3	core	none	3–9	1,484	163	8	8.1	0.40	13
contraceptive [minority: 2]	high	none	3–9	1,473	333	9	3.4	0.79	2
kr-vs-k-one_vs_fifteen	low	none	>9	2,244	78	6	27.8	0.00	1
shuttle-c0-vs-c4	core	none	>9	1,829	123	9	13.9	0.01	7
yeast-0-2-5-6_vs_3-7-8-9	high	none	>9	1,004	99	8	9.1	0.57	4
heart	core+low	S	≤1.5	270	120	13	1.2	0.33	4
sonar	high	S	≤1.5	208	97	60	1.1	0.40	1
wine [minority: 1]	low	S	1.5–3	178	59	13	2.0	0.00	1
glass0	core	S	1.5–3	214	70	9	2.1	0.44	7
planning-relax	high	S	1.5–3	182	52	12	2.5	0.85	1
musk	low	S	3–9	6,598	1,017	167	5.5	0.01	2
ecoli3	core	S	3–9	336	35	7	8.6	0.69	10
hepatitis (PMLB, N=155)	high	S	3–9	155	32	19	3.8	0.75	2
dermatology-6	low	S	>9	358	20	34	16.9	0.00	2
yeast5	core	S	>9	1,484	44	8	32.7	0.66	13
cleveland-0_vs_4	high	S	>9	177	13	13	12.6	1.00	4
shoppers	core+low	O	3–9	12,330	1,908	17	5.5	0.62	4
pima2	high	O	3–9	634	134	8	3.7	0.80	1
mammography	core+low	O	>9	11,183	260	6	42.0	0.54	5
balance [minority: B]	high	O	>9	625	49	4	11.8	0.88	2
heart6	low	S+O	3–9	170	20	13	7.5	0.65	1
breast_tissue4	core+high	S+O	3–9	79	9	9	7.8	0.89	1
ecoli-0-1-3-7_vs_2-6	low	S+O	>9	281	7	7	39.1	0.29	7
yeast4	core	S+O	>9	1,484	51	8	28.1	0.86	15
ijcnn1	high	S+O	>9	4,858	5	22	970.6	1.00	1
haberman2	core+low+high	D+O	3–9	265	40	3	5.6	0.70	2
promotion	core+low+high	D+O	>9	23,815	1,634	4	13.6	0.77	1
breast.cancer	core+low+high	D+S	1.5–3	277	81	9	2.4	0.57	3
iris4	core+low+high	D+S	3–9	112	12	4	8.3	0.75	1
iris6	core+low+high	D+S	>9	108	8	4	12.5	0.50	1
heart-disease-processed-cleveland [minority: 4]	low	D+S+O	3–9	303	35	13	7.7	0.71	1
fertility-diagnosis	core	D+S+O	3–9	100	12	9	7.3	0.75	1
liver_disorders6	high	D+S+O	3–9	224	24	6	8.3	0.79	1
flare [minority: 5]	low	D+S+O	>9	1,389	51	12	26.2	0.59	1
glass2 (KEEL(keel_ds), N=214)	core	D+S+O	>9	214	17	9	11.6	0.65	9
pima6	high	D+S+O	>9	544	44	8	11.4	0.80	1

Table 8. Comparisons of the Proposed Method with Plain SMOTE by Metric (29 Papers). A row is one dataset and one metric in one paper; printed values are compared in the direction of the metric, equal printed values are ties, and n.c. counts values not printed as numbers. Scores are not comparable across papers.

Metric	Papers	Rows	Proposed wins	Ties	SMOTE wins	n.c.
AUC	19	403	309	30	64	0
G-mean	19	380	283	29	64	4
F-measure	18	287	217	16	50	4
Recall (sensitivity)	12	193	130	21	42	0
Precision	7	113	62	5	46	0
Accuracy	4	69	62	1	6	0
MCC	3	52	40	0	8	4
AUC-PR	1	7	3	0	0	4
Specificity	1	7	1	0	6	0
All metrics	29	1,511	1,107	102	286	16

Table 9. Comparisons with Plain SMOTE by How the Dataset Was Obtained (All Metrics, 29 Papers). Win share is the share of wins among decided comparisons (ties left out), with a 95% Wilson interval. Sixteen comparisons not printed as numbers are left out.

How obtained	Datasets	Papers	Rows	Wins	Ties	Losses	Win share (%)
Original	143	25	422	327	10	85	79.4 (75.2–83.0)
Repository-defined	119	9	315	220	44	51	81.2 (76.1–85.4)
Modified	143	14	495	356	20	119	74.9 (70.9–78.6)
Synthetic	6	2	18	17	0	1	94.4 (74.2–99.0)
Cannot be told	105	13	234	184	24	26	87.6 (82.5–91.4)
Not matched to a dataset	4	3	11	3	4	4	42.9 (15.8–75.0)
All	520	29	1,495	1,107	102	286	79.5 (77.3–81.5)

Table 10. Granularity of the Results Each Paper Reports (73 Papers)

How results are reported	Papers
Per-dataset table	47
Per-dataset, partial (some datasets, metrics or classifiers only)	8
Plots only	5
Averages over all datasets only	5
Ranks only	3
No comparison with other methods	5